Omjee R Giri

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PROFESSIONAL SUMMARY

AI/ML engineering student (B.Tech AI&ML, BIT Mesra; BS Data Science, IIT Madras) building RAG, LLM and NLP systems with hybrid retrieval, reranking, citation validation and automated evaluation. Undergraduate research assistant, GSSoC 2026 contributor and Codeforces Specialist (Max - 1491), seeking AI/ML and Software Engineering internships.

EDUCATION

Birla Institute of Technology, Mesra Aug 2024 – Present
B.Tech, Artificial Intelligence and Machine Learning CGPA: 7.6/10

Indian Institute of Technology, Madras Jan 2025 – Present
BS, Data Science and Applications CGPA: 7.0/10

WORK EXPERIENCE

Undergraduate Research Assistant – BIT Mesra, Dept. of Computer Science Jan 2026 – Present
- Design and build Legal Lens, an evidence-grounded legal research system for Indian law, owning its backend and retrieval architecture (details under Projects).
- Set up the evaluation harness (Recall@K, MRR, nDCG, citation validity) used to compare chunking and reranking strategies.

Open-Source Contributor – GirlScript Summer of Code (GSSoC) 2026 May 2026 – August 2026
- Merged 7 PRs into Nyay Setu, a digital judiciary platform | github.com/viru0909-dev/nyay-setu-working
- Added provider retry and fallback to the NLP orchestrator, and a citation-verification layer with BNS/BNSS section-range rules and 40+ Pytest cases.
- Added streaming token responses, caching for repeated NLP requests, and IP-based rate limiting on authentication endpoints.

PROJECTS

Legal Lens: Evidence-Grounded Legal Research for Indian Law github.com/TheOneAbovELL/legal-lens-backend
- Built a 16-node LangGraph RAG pipeline that routes each query to one of 3 retrieval profiles via a rule-based complexity score, fusing BGE-M3 dense, BM25-style sparse, metadata and Neo4j retrieval on Qdrant with weighted RRF.
- Added an evidence gate that skips the LLM on weak evidence and citation validation that rejects unknown citation IDs; model calls use retries, provider fallback and a circuit breaker.
- On an 18-query labelled set, a lexical reranker matched a cross-encoder at about a quarter of the latency, so it became the default; 160+ offline tests, Playwright E2E and CI.

Tech Stack: Python, FastAPI, LangGraph, Qdrant, BGE-M3, Groq, Neo4j, React, TypeScript, Docker, GitHub Actions

qEEG Subtyping: Multi-Branch Deep Learning on EEG *Research Prototype (in progress)*
- Built 19-channel EEG preprocessing (filtering, ICA, re-referencing, epoching) and spectral, connectivity and nonlinear feature extraction.
- Designing a three-branch classifier (1D-CNN for local patterns, GCN over measured channel coherence, Transformer for long-range dependencies) fused into a six-class output.

Tech Stack: Python, PyTorch, MNE-Python, NumPy, SciPy

ARGUS: Quantitative Research Backend github.com/TheOneAbovELL/ARGUS
- Built an equity research backend where no metric comes from an LLM: pure, unit-tested functions compute return, volatility, Sharpe, beta and max drawdown for assets and portfolios.
- Runs 4 rule-based agents concurrently with asyncio, plus a ChromaDB retrieval layer that returns page-level provenance; 56+ offline tests.

Tech Stack: Python, FastAPI, AsyncIO, Pydantic, ChromaDB, yfinance, Next.js

Semantic Product Recommendation Engine github.com/TheOneAbovELL/AMAZON-REC-ENGINE
- Built retrieve-then-rank recommendations with all-MiniLM-L6-v2 embeddings and FAISS over 50,000+ products (sub-second retrieval), re-ranked on five signals including similarity, rating and budget fit.

Tech Stack: Python, Sentence Transformers, FAISS, Pandas, scikit-learn, Streamlit | Live: amazon-rec-engine.vercel.app

SatQuery AI: Satellite Imagery Q&A Assistant (SIH 2026) github.com/TheOneAbovELL/SATQUERY
- Built a vision-language assistant for ISRO's SIH problem statement 26167 that answers natural-language questions over optical, multispectral and SAR imagery using Qwen2-VL-2B-Instruct with a remote-sensing LoRA adapter.
- Routed each query to deterministic geospatial tools for the measurements, so the model interprets evidence but never invents numbers; every answer carries an auditable execution trace.

Tech Stack: Python, FastAPI, Qwen2-VL, PEFT (LoRA), Rasterio, Shapely, NumPy, SciPy, Next.js

ACHIEVEMENTS

- Codeforces Specialist – rating 1491 (C++) | codeforces.com/profile/The0NEAboveALL
- GirlScript Summer of Code 2026 – global rank 1,065 of 70,000+ registered contributors, 7 PRs merged

TECHNICAL SKILLS

Languages and Core CS: Python, C++, C, TypeScript, JavaScript, SQL; DSA, OOP, DBMS, OS, Computer Networks
AI / ML: PyTorch, scikit-learn, Hugging Face, LangGraph, RAG, hybrid retrieval, reranking, vision-language models, LoRA (PEFT)
Data and Geospatial: NumPy, Pandas, SciPy, MNE-Python, Rasterio, Shapely, Qdrant, ChromaDB, FAISS, Neo4j, MySQL
Backend and DevOps: FastAPI, Pydantic, AsyncIO, SSE, WebSockets, React, Next.js, Docker, GitHub Actions, Pytest, Playwright, Git